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Abstract

This case study presents a machine-learning system that predicts whether a visitor to an electronics e-commerce store is likely to purchase a specific product, and uses that prediction to drive a personalised product-recommendation engine. The system is built entirely on a real, uploaded click-stream dataset of 885,129 view, cart and purchase events recorded between 24 September 2020 and 28 February 2021, covering 407,283 users, 53,453 products, 718 category identifiers (107 descriptive category codes) and 999 brands. No product names, descriptions, ratings or cancellation events exist in the data, so every design decision in this report is made to fit the schema that is actually present rather than an idealized one.

After cleaning and leakage-safe feature engineering, 631,353 user-product interaction samples were constructed. Each sample corresponds to the first time a user viewed a particular product in a session, and is labelled 1 if that same user purchased that same product within the following 7 days (a 5.01% positive rate) and 0 otherwise. Two classifiers, Logistic Regression and HistGradientBoosting, were trained on a strict time-based split with a purge gap between training, validation and test periods, and were compared on the validation period using ROC-AUC and PR-AUC. HistGradientBoosting was selected (validation PR-AUC 0.163 against 0.135 for Logistic Regression) and evaluated once on a held-out February test period, reaching ROC-AUC 0.775, PR-AUC 0.170, accuracy 86.9%, precision 17.7%, recall 37.2% and F1-score 0.240. The trained model is then used to score candidate products for any user and generate a ranked, interactive Top-N recommendation list driven by the user's selected interest category, keyword and price ceiling, demonstrating that genuine behavioral signals, rather than fabricated data, can support personalisation even on a dataset with no textual product information.

Introduction

Background

Modern e-commerce platforms list thousands of products, and a typical visitor views only a small fraction of the catalogue before leaving, adding an item to the cart, or completing a purchase. Traditional recommendation approaches, such as collaborative filtering and matrix factorisation, mainly rely on long-run patterns of similarity between users or between items and often respond slowly to what a specific visitor is doing right now. A system that instead reads the customer's recent views and cart activity, estimates how likely they are to buy a specific product in the near future, and ranks candidate products by that estimate can react to purchase intent as it forms during a single browsing session.

Motivation for the Project

The motivation for this project is to build a purchase-prediction and recommendation pipeline that is fully defensible against the dataset it is trained on: every feature, every category and every recommended product must be traceable to a column that genuinely exists in the uploaded events

file. This constraint was treated as a design requirement rather than a limitation, because it forces the pipeline to demonstrate that useful, personalised predictions can be produced from ordinary clickstream signals (views, cart additions, price, category and brand) without inventing product descriptions, ratings or user profiles that the store never actually collected.

Overview of the Work Carried Out

- Inspected the uploaded ZIP archive and the single events.csv file inside it, confirming its exact schema, size, and date range before writing any modelling code.
- Performed data cleaning: removed exact duplicate rows and rows with a missing session identifier, and verified that every event type, timestamp and price was valid.
- Carried out exploratory data analysis (EDA) covering event-type mix, missing-value patterns, user and product activity, category and brand popularity, price distribution and time-of-day/day-of-week/monthly activity, in a dedicated EDA notebook.
- Defined a leakage-safe binary purchase-prediction target and engineered 33 behavioural features computed only from information available before the prediction point.
- Trained and compared a Logistic Regression baseline and a HistGradientBoosting classifier on a time-based train/validation/test split, and evaluated the selected model with a full metric suite.
- Built a model-driven recommender that scores real candidate products for a user and an interactive command-line demo in which the user's choices change the recommendations shown.

Scope and Responsible-Use Statement

The scope of this case study is limited to the single uploaded electronics-store events dataset; no external product catalogue, pricing data or personal information was introduced. The user_id and user_session values in the dataset are anonymised numeric and alphanumeric identifiers, and no personally identifiable information is processed, stored or displayed at any stage of the pipeline. The purchase-probability scores produced by the model are statistical estimates derived from historical behaviour, not guarantees of future action, and are intended only for this academic demonstration. The system is not deployed against real customers, does not make autonomous purchasing or pricing decisions, and would require further fairness, privacy and robustness review before any production use.

Problem Statement

Case-Study Scenario

Consider a customer browsing the electronics store. During a single session, the customer views several products in the computers.notebook category, repeatedly returns to one particular laptop, adds it to the cart, and then also looks at a few accessories from a related category. The customer

does not complete the purchase immediately. Using only the events the store has actually recorded, the system must decide which products to show this customer next: continuing to promote generic best-sellers would ignore the clear signal that this visitor is specifically interested in laptops and related accessories, while a system that reads the session correctly should prioritise similar and complementary products and estimate how likely the customer is to buy them soon.

Central Case-Study Question

Given only a user's real product views, cart additions and purchase history recorded up to a given moment, can a machine-learning model estimate the probability that the user will purchase a specific product within a short, clearly defined future time window, and can that estimate be used to rank real products from the catalogue into a personalised recommendation list?

Formal Problem Definition

Purchase prediction is formulated as binary classification. For every (user, product) pair, a decision point is fixed at the first time that user viewed that product within a session. Using only events strictly before this decision point, a feature vector is built describing the user's history, the product's and its category's and brand's recent popularity, and relative price. The target label is 1 if the same user purchased the same product within 7 days after the decision point, and 0 otherwise. The recommendation problem is then: for a given user and a chosen candidate set of real products, rank the candidates by the trained model's predicted purchase probability and return the highest-ranked items.

Objectives that Guide the Study

- Use only columns, event types and values that genuinely exist in the uploaded dataset, with no invented product names, categories, brands or outcomes.
- Construct a binary purchase target that is defensible against target leakage, using strictly historical information for every feature.
- Engineer features that summarise user history, product/category/brand popularity and price in a way that is computable in real time for a live visitor.
- Compare at least two classical machine-learning models and select one on validation-period performance rather than on the test period.
- Evaluate the final model with accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC and a confusion matrix, since the purchase class is a small minority of all interactions.
- Connect the trained model directly to a personalised, explainable recommendation pipeline rather than a hard-coded list of products.
- Build a simple interactive demonstration in which a real user's choice of category, keyword and price ceiling visibly changes the recommended products.

Assumptions and Limitations

- The dataset contains only view, cart and purchase events; there is no returns, cancellation or rating information, so only completed purchases could be used as a positive outcome.
- category_code is missing for 26.69% of events and brand for 23.99%; these were kept as an explicit "unknown" value rather than being dropped or guessed.
- The dataset has no product name, description or free-text field, so recommendations and the interactive demo are built around category, brand and price rather than natural-language search.
- Only 5.23% of the 407,283 users in the dataset ever complete a purchase, so the prediction target is strongly imbalanced and precision/recall trade-offs must be read with that base rate in mind.
- The data spans about five months from a single store; behaviour outside that period, from other stores, or from users with very little history (the cold-start case) is not represented.
- Purchase probability estimates reflect correlation in historical behaviour, not a causal explanation of why a customer buys a product.

Dataset and Exploratory Data Analysis

About the Dataset

The dataset is a single file, events.csv, inside the uploaded archive, containing 885,129 rows and 9 columns: event_time, event_type, product_id, category_id, category_code, brand, price, user_id and user_session. It records 793,748 view events (89.67%), 54,035 cart events (6.11%) and 37,346 purchase events (4.22%) between 24 September 2020 11:57:06 UTC and 28 February 2021 23:59:09 UTC. It covers 407,283 unique users, 53,453 unique products, 718 category identifiers grouped into 107 descriptive category codes (for example computers.components.videocards, electronics.telephone and appliances.environment.vacuum), 999 brands and 490,398 browsing sessions. Prices range from 0.22 to 64,771.06, with a mean of 146.33 and a median of 65.71.

Purpose of EDA

Exploratory data analysis was carried out before any modelling decision in order to understand what the dataset genuinely supports: how complete each column is, how active users and products are, how the event funnel narrows from view to cart to purchase, how price relates to event type, and how activity varies over time. These findings directly shaped the choice of a 7-day, first-view-anchored prediction target and the decision to represent missing category and brand values explicitly rather than discarding a quarter of the data.

Preparation Steps Actually Performed

- Loaded events.csv directly from the uploaded ZIP archive and parsed event_time to a timezone-aware timestamp.
- Removed 652 exact duplicate rows and 165 rows with a missing user_session value, leaving 884,312 usable rows.
- Confirmed there were no unparseable timestamps, no event types outside view/cart/purchase, and no non-positive or missing prices, so no further row-level repair was needed.
- Verified that every product has exactly one associated price and one associated category in the data, so product attributes could be treated as a static catalogue when building features.
- Filled missing category_code and brand with an explicit "unknown" category rather than dropping the affected rows, preserving the view/cart/purchase signal they still carry.

EDA Exploration - Implementation

The accompanying EDA notebook (PURCHASE_PREDICTION__PERSONALIZED_PRODUCT_EDA.ipynb) inspects the dataset shape, column types and missing-value counts; plots the event-type distribution and a missing-values bar chart; counts unique users, products, categories, brands and sessions; profiles customer and product activity (most active customer: 572 events, most-viewed product: 1,821,813 with 14,552 interactions); ranks the top 15 categories, category codes and brands by interaction count; examines the price distribution overall and split by event type; and breaks activity down by hour of day, day of week and month, as well as by number of events per session. It finishes by labelling every user as Exploring, Considering or Purchased from a simple rule over their own event counts, purely as a descriptive summary rather than a modelling feature.

Summary of EDA Findings

- The purchase funnel is heavily top-weighted: 89.67% of all events are views, 6.11% are cart additions and only 4.22% are purchases.
- Only 21,304 of 407,283 users (5.23%) completed at least one purchase in the five-month window; the simple engagement labelling put 369,083 users at "Exploring", 16,896 at "Considering" and 21,304 at "Purchased".
- computers.components.videocards is by far the most viewed category code (116,611 interactions), followed by electronics.telephone (84,308) and computers.peripherals.printer (43,193).
- Activity is fairly stable across the week (117,494 to 133,738 events per weekday) and rises through the morning to a plateau between roughly 09:00 and 18:00 UTC before tapering off overnight.

- category_code and brand are missing for roughly a quarter of events each, which is why both were kept as an explicit "unknown" category rather than removed.
- The average price of carted items (159.64) is higher than that of purchased items (137.24) and viewed items (145.85), suggesting price-sensitivity plays a role between adding an item to the cart and actually buying it.

Algorithm Selection and Implementation

Target construction. For every (user, session, product) combination, the first view row is taken as the decision point. A forward, non-overlapping join locates the next purchase of the same (user, product) pair strictly after that point in the time-ordered log; the sample is labelled 1 if that purchase occurs within a 7-day horizon, and 0 otherwise. This produced 631,353 labelled samples with a 5.01% positive rate. Because every feature below is computed from events that occurred strictly before the decision point, the model cannot see any information from the labelling window itself, which avoids target leakage.

Feature engineering. 33 features were engineered, grouped into: product popularity (p_views, p_carts, p_purchases and the smoothed $p_purchase_rate / p_cart_rate$), category popularity (c_views, c_carts, c_purchases, c_purchase_rate, c_cart_rate), brand popularity (b_views, b_carts, b_purchases, b_purchase_rate, b_cart_rate), price (log_price, price_vs_category, price_vs_user_mean), user history (u_views, u_carts, u_purchases, u_sessions, u_days_since_first, u_hours_since_last), user-category and user-product affinity (uc_views, uc_carts, uc_purchases, uc_view_share, up_views, up_carts, up_purchases) and two catalogue attributes (top_category, has_brand). All popularity and history statistics are cumulative counts computed only from events before the decision point.

Train / validation / test split. The 631,353 samples were split strictly by time, with purge gaps of at least 7 days around each boundary so that no label window spans two periods: training used 361,941 samples from 24 September to 24 December 2020 (4.65% positive), validation used 87,691 samples from 1 to 21 January 2021 (5.38% positive) and the final, untouched test set used 90,212 samples from 1 to 21 February 2021 (5.55% positive).

Model comparison and class imbalance. A Logistic Regression baseline and a HistGradientBoostingClassifier were both trained with balanced class weighting to offset the roughly 19:1 ratio of negative to positive samples, and the decision threshold was tuned on the validation period only, never on the test period. On validation, Logistic Regression reached ROC-AUC 0.7571 and PR-AUC 0.1345, while HistGradientBoosting reached ROC-AUC 0.7725 and PR-AUC 0.1627. HistGradientBoosting was selected as the final model because it gave the better validation PR-AUC, the more informative metric here given how rare the positive class is; its decision threshold on the raw score was tuned to 0.705.

Test-set evaluation. On the held-out February test period, the final HistGradientBoosting model reached an accuracy of 86.93%, precision of 17.72%, recall of 37.15%, F1-score of 0.2399, ROC-AUC of 0.7747 and PR-AUC of 0.1701, with a Brier score of 0.0492. The confusion matrix was 76,558 true negatives, 8,644 false positives, 3,149 false negatives and 1,861 true positives. Accuracy alone is misleading here because simply predicting "no purchase" for every sample would already score about 94.5% accuracy; ROC-AUC, PR-AUC, recall and F1 are reported precisely because the purchase class is rare, and they show the model recovers roughly 37% of actual purchases while ranking purchasers well above non-purchasers overall (ROC-AUC 0.77).

Recommendation system. The trained model is wrapped in a Recommender that holds the full product catalogue (53,452 products across 407,237 users). Given a user and a set of candidate products built from the real category, brand and price columns, it computes the same 33 features for every (user, candidate) pair and ranks the candidates by the model's predicted purchase probability, so every recommendation is produced by the model itself rather than being hard-coded. In the interactive demo, a user who selected the appliances category, the keyword "appliances" and a maximum price of 200 received ten ranked recommendations drawn only from matching real products, for example product 914374 (appliances.environment.vacuum, brand pitatel, price 38.10, purchase probability 13.9%) down to product 1434012 (appliances.kitchen.grill, brand bosch, price 119.70, purchase probability 9.6%), with every product, category, brand and price taken directly from the dataset and every probability produced by the trained classifier.

Conclusion

This case study built a Customer Purchase Prediction and Personalised Product Recommendation System entirely from a real, inspected electronics-store events dataset, without inventing any product, category, brand or behavioural information. A leakage-safe, time-based pipeline turned raw view/cart/purchase logs into a defensible 7-day purchase target, 33 behavioural features, and a HistGradientBoosting classifier that outperformed a Logistic Regression baseline on validation PR-AUC (0.163 against 0.135) and generalised to an unseen test period with ROC-AUC 0.775 and recall of 37% on a minority class that makes up only about 5% of all interactions. The trained model was connected directly to a recommendation engine and an interactive demo, confirming that a user's own choices of category, keyword and price genuinely change the recommended products and their predicted purchase probabilities. The main limitations are the short, five-month data window, the absence of product names, descriptions or ratings, and the inherent difficulty of predicting a rare event, which keeps precision modest even though the model ranks purchasers well above non-purchasers. Future work could extend the feature set with richer session-level signals, explore gradient-boosted alternatives such as XGBoost, and evaluate the recommender online rather than purely offline.

Signature of the Team members